

Braided Operads

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Outline

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Operads

Definition 2.1

An *operad* $\mathcal{O}$ in $\mathcal{C}$ consists of

- a family of objects $(\mathcal{O}(j))_{j \in \mathbb{N}}$
- a unit morphism $\eta: k \rightarrow \mathcal{O}(1)$
- a composition morphism

$$\gamma: \mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \cdots \otimes \mathcal{O}(j_k) \rightarrow \mathcal{O}(j),$$

where $\sum_{1 \leq i \leq k} j_i = j$

- an action of S_n on $\mathcal{O}(n)$

such that suitable associativity, equivariance, unitality constraints hold.

Definition 2.2

Let $P: \mathcal{C} \rightarrow \mathcal{D}$ be a functor of arbitrary categories and $f: x \rightarrow y$ a morphism in $\mathcal{C}$. We call f *cocartesian* if for any morphism $g: y \rightarrow z$ in $\mathcal{C}$ and $\omega: P(x) \rightarrow P(z)$ in $\mathcal{D}$ with $\omega \circ P(f) = P(g)$, there is a unique morphism $\omega': x \rightarrow z$ such that $P(\omega') = \omega$ and $\omega' \circ f = g$. In a diagram this results in the following:

$$\begin{array}{ccc}
 y & \xrightarrow{g} & z \\
 & \searrow f & \uparrow \exists! \omega' \\
 & & x
 \end{array}
 \quad \xrightarrow{P} \quad
 \begin{array}{ccc}
 P(y) & \xrightarrow{P(g)} & P(z) \\
 & \searrow P(f) & \uparrow \omega \\
 & & P(x)
 \end{array}$$

Definition 2.3

A *colored operad* $\mathcal{O}$ consists of:

- (i) A collection $\{X, Y, Z, \dots\}$ which we call *colors*.
- (ii) For finite collection of colors $\{X_i\}_{i \in I}$, color Y , a set: $\text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in I}, Y)$ called the set of morphisms.
- (iii) For a finite map $I \rightarrow J$ with fibers $I_j, j \in J$, collections of colors $\{X_i\}_{i \in I}$, $\{Y_j\}_{j \in J}$, color Z , a composition map:

$$\prod_{j \in J} \text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in I_j}, Y_j) \times \text{Mul}_{\mathcal{O}}(\{Y_j\}_{j \in J}, Z) \rightarrow \text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in I}, Z).$$

- (iv) A collection of morphisms

$$\{\text{Id}_X \in \text{Mul}_{\mathcal{O}}(\{X\}, X)\}_{X \in \mathcal{O}}$$

called the identity morphisms.

The data needs to fulfill suitable associativity and unitality conditions.

Example 2.4

Let $\mathcal{C}$ be a symmetric monoidal category. By setting for a finite family of objects $\{X_i\}_{i \in I}$ and object Y in $\mathcal{C}$:

$$\text{Mul}_{\mathcal{C}}(\{X_i\}_{i \in I}, Y) := \text{Hom}(\otimes_{i \in I} X_i, Y),$$

we can define a colored operad $\mathcal{C}$. The composition is given by

$$\prod_{j \in J} \text{Hom}(\otimes_{i \in I_j} X_i, Y_j) \times \text{Hom}(\otimes_{j \in J} Y_j, Z) \rightarrow \text{Hom}(\otimes_{i \in I} X_i, Z)$$

$$((f_j)_{j \in J}, g) \mapsto g \circ (\otimes_{j \in J} f_j)$$

with $\{\text{Id}_X\}_{X \in \mathcal{C}}$ being the unit and associativity holds since for

$(f_j: \otimes_{i \in I_j} X_i \rightarrow Y_j)_{j \in J}$, $(g_k: \otimes_{j \in J_k} Y_j \rightarrow W_k)_{k \in K}$, $h: \otimes_{k \in K} W_k \rightarrow Z$, we have

$$h \circ \bigotimes_{k \in K} \left(g_k \circ \bigotimes_{j \in J_k} f_j \right) = h \circ \bigotimes_{k \in K} g_k \circ \bigotimes_{j \in J} f_j.$$